2.7.23

EE25BTECH11011 - Navya

Question:
For what value of k ($k > 0$) is the area of the triangle with vertices $(-2, 6)$, $(2k, 4)$ and $(2k + 1, 10)$ equal to 52 sq. units?

Solution:
Vertices of triangle are

$$\mathbf{A} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} \tag{0.1}$$

$$\mathbf{B} = \begin{pmatrix} 2k \\ 4 \end{pmatrix} \tag{0.2}$$

$$\mathbf{C} = \begin{pmatrix} 2k + 1 \\ 10 \end{pmatrix} \tag{0.3}$$

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} -2 - 2k \\ 6 - 4 \end{pmatrix} = \begin{pmatrix} -2 - 2k \\ 2 \end{pmatrix} \tag{0.4}$$

$$\mathbf{A} - \mathbf{C} = \begin{pmatrix} -2 - (2k + 1) \\ 6 - 10 \end{pmatrix} = \begin{pmatrix} -2k - 3 \\ -4 \end{pmatrix} \tag{0.5}$$

$$(\triangle ABC) = \frac{1}{2} \left\| (\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C}) \right\| \tag{0.6}$$

$$(\triangle ABC) = \frac{1}{2} \left\| \begin{pmatrix} -2 - 2k \\ 2 \end{pmatrix} \times \begin{pmatrix} -2k - 3 \\ -4 \end{pmatrix} \right\| = \frac{1}{2} |12k + 14| \tag{0.7}$$

Given area of triangle = 52 sq.units,

$$\frac{1}{2} |12k + 14| = 52 \tag{0.8}$$

$$|12k + 14| = 104. \tag{0.9}$$

Now consider the two cases from the absolute value,

Case 1:

$$12k + 14 = 104 \tag{0.10}$$
$$12k = 90 \tag{0.11}$$
$$k = \frac{90}{12} = \frac{15}{2}. \tag{0.12}$$

Case 2:

$$12k + 14 = -104 \tag{0.13}$$
$$12k = -118 \tag{0.14}$$
$$k = -\frac{118}{12} = -\frac{59}{6}. \tag{0.15}$$

Because the problem requires $k > 0$, we discard the negative solution.

$$\boxed{k = \frac{15}{2}}$$